

Applied MSc in Data Science & AI

Applied MSc in Data Engineering & AI

Deep Learning with Python

Course Project (2026)

Instructor: Benoit Mialet

Purpose of the project:

The goal of this project is to practice deep learning on a realistic but manageable problem.

You will:

- Work in groups (2-4 students).
- Explore one NLP or Computer Vision task
- Use PyTorch and/or Hugging Face
- Train or fine-tune at least one open-source (or handmade) model
- Work with one or more public dataset(s)
- Perform a small number of meaningful experiments
- Clearly justify your choices
- **Prioritize scientific reasoning over engineering complexity**

The objective is not to obtain the best score, but to demonstrate a sound experimental methodology.



Experimental workflow:

- Choose one task (NLP or CV) from <https://huggingface.co/tasks>
- Use at least one open-source dataset from <https://huggingface.co/datasets>
- Use at least one (open-source pretrained from <https://huggingface.co/models> or from scratch) model (small ones are recommended for Google Collab notebooks)
- Write train & evaluation methods
- Evaluate baseline model(s)
- **Option 1:** perform 2–3 controlled improvements (e.g. hyperparameter tuning, data augmentation, different optimizer, freezing vs. unfreezing layers, different datasets with same model, different models with same dataset, different preprocessing, etc.)
- **Option 2:** compare model performance with different transfer learning strategies (fine tuning, freezing, distillation, etc.)
- **Option 3:** any other since it involves deep learning...
- Analyze and discuss the results

What you have to produce:

You must deliver the following components:

A. Technical Project Summary (Mandatory)

This is the main element I will grade.

Format:

- PDF, maximum 4-8 pages
- Use concise text and bullet points rather than long paragraphs.

The document should contain:

Project Overview

- Project title
- Problem addressed and why it is interesting

Method

- **Your overall strategy**



- Model(s) used
- **Description of the dataset(s) + short EDA** (only most important aspects)
- **Data Preprocessing steps**
- Training setup: train/val/test split, metrics, hyperparameters
- Hyperparameters tuning

Results

- Comment tables, figures and metrics
- Comparisons if relevant (e.g., baseline vs fine-tuned model)

Discussion

- What do the results mean?
- What worked / what didn't?
- Any unexpected outcomes?
- Limitations
- **How would you improve with more time/data/resources/other methods?**

Short conclusion

- What is the final takeaway of your project?

Annexes (optional)

- Extra figures, tables, logs

B. Code (Mandatory)

- You must provide (notebooks or python modules):
 - **EDA**
 - **Data Preprocessing pipeline**
 - **Model Training pipeline**
 - **Model Evaluation**
- **Method must be reproducible**, write or store requirements & settings
- Keep cells outputs or graph files so that you comment them in result section
- **Don't explain things twice in code and in project summary**, to save effort

D. Extras (Optional, not graded, but nice for your CV)

- A GitHub repository (also better for team work)
- **Don't attach models to the project.**

Summary of What Matters Most

- A clear scientific approach
- Your own reasoning. Explanations matter more than performance
- Reproducible notebooks
- Concise document
- Use of PyTorch and/or Hugging Face
- Careful use of generated code

You may request a short meeting with the instructor (optional) to get feedback, guidance, or help defining the purpose of your project.